

Pattern-sensitive Epilepsy: Electroclinical Characteristics, Natural History, and Delineation of the Epileptic Syndrome

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Summary: *Purpose:* To elucidate the electroclinical features and long-term outcome of patients with pattern-sensitive epilepsy.

Methods: We reviewed the clinical and electroencephalographic (EEG) findings of 73 (43 female and 30 male) patients in whom pattern-sensitive epilepsy was diagnosed at Mayo Clinic (Rochester, Minnesota, U.S.A.) from 1950 through 1999. We contacted patients and their relatives by letter or telephone to obtain the latest seizure and quality-of-life outcomes.

Results: The median age at onset of seizures was 12.8 years (range, 0.6–32.9 years). Most patients had absence, myoclonic, or generalized tonic–clonic seizures. Interictal epileptiform discharges in the EEG were detected in 61 (83.6%) patients and were generalized in 54 (74%). Paroxysmal epileptiform discharges in the EEG elicited with standard patterns were all generalized in two thirds of patients but were restricted to the posterior head region in one-third. Eight (11%) patients did not exhibit photosensitivity. Television was the most common precipitant [30 patients (41%)]. Twenty-nine patients gave a clear history of one or more seizures precipitated while viewing environmental patterns such as window screens, garments, tablecloths, and ceiling tiles; the rest of the patients admitted that they preferred to avoid looking at patterned objects because these objects made them uncomfort-

able. The electroclinical features suggested juvenile myoclonic epilepsy in 14 patients, progressive myoclonus epilepsy in three, progressive familial cerebellar ataxia with myoclonus in two, and severe myoclonic epilepsy of infancy in one. During a median follow-up period of 15.7 years, 25 (45.5%) of 55 patients who were followed up for ≥ 5 years achieved complete seizure remission. The median age at remission was 24.4 years. The absence of progressive neurologic disease was correlated significantly with remission; a family history of seizures showed a trend in favor of remission. More than two thirds of the patients did not consider the seizures an impediment to their family life or to educational and occupational achievements.

Conclusions: Although pattern sensitivity as a trait occurs in various epileptic syndromes, pattern-sensitive epilepsy is a readily distinguishable subtype of the visually provoked reflex epilepsies. In our opinion, the location and extent of the excitable region or regions within the visual cortex concerned with different attributes of visual function dictate susceptibility to a specific trigger (intermittent light, pattern, or color) or closely related multiple triggers and the resultant electroclinical phenomenon.

Key Words: Pattern-sensitive epilepsy—Pattern sensitivity—Photosensitivity—Reflex epilepsy—Television epilepsy.

It has long been recognized that in certain individuals, epileptic seizures can be precipitated by a wide variety of external stimuli (1,2). Visual stimuli are by far the most common trigger of these reflex epilepsies (2,3). Either simple visual stimuli such as light or patterns (4,5) or complex visual excitations such as television or video games may trigger visually induced seizures in subjects with television- or video game-induced seizures (6,7).

Light-induced seizures became widely recognized after the introduction of intermittent photic stimulation during electroencephalographic (EEG) recording (8). However, it was not until 1953 that the first report of a patient with visual pattern sensitivity appeared (9). The patient was a 6-year-old boy who had spells of blinking and unresponsiveness that were brought on when he would seek out and actively gaze at window screens or finely patterned fabrics such as his father's necktie, corduroy jacket, or poplin storm coat. Paroxysmal epileptiform EEG abnormalities and clinical seizures could be induced in the EEG laboratory when the boy would look at finely patterned clothing, a window screen, and pictures. He also was markedly sensitive to intermittent photic stimulation. Further studies demonstrated that one of the most effective triggering stimuli was a pattern of parallel black and white lines

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aligned vertically, but the same pattern was ineffective when the lines were aligned horizontally (2,3,10).

During the past 50 years, many reports of individual patients (11,12) and series of patients (13–16) with pattern sensitivity and comprehensive reviews (4,5) of the subject have appeared. These studies have elaborated on the demographic and clinical features of subjects with pattern sensitivity, the stimuli most effective in provoking pattern sensitivity in the environment and in the laboratory, electrophysiologic mechanisms of pattern sensitivity, and the interrelation of pattern sensitivity with other visually induced seizures such as light, television, video games, and reading.

A close association exists between sensitivity to intermittent photic stimulation (photosensitivity) and sensitivity to visual pattern stimulation (pattern sensitivity). In one third to two thirds of patients with photosensitivity, static or oscillating patterns induce epileptiform abnormalities in the EEG, depending on the patient population studied and the method of testing (4,5). Pattern sensitivity in isolation, without photosensitivity, is believed to be almost nonexistent (17). Because most EEG laboratories do not routinely perform pattern-sensitivity testing and conduct it only in subjects who exhibit an abnormal response to intermittent photic stimulation, it is uncertain whether the documented association between photosensitivity and pattern sensitivity is real or a result of selection bias.

Whereas pattern sensitivity is a trait characterized by epileptiform abnormalities demonstrated in the EEG laboratory, pattern-sensitive epilepsy is a condition in which environmental patterns are the sole factor or one of the major seizure-triggering factors. Many investigators consider pattern-sensitive epilepsy an extension of photosensitive epilepsy and, therefore, have described the demographic and clinical characteristics of patients with pattern-sensitive seizures by ascertaining them from patients with photosensitivity (4,18). Consequently, in most of the reported series, pattern sensitivity was not the probable cause of epilepsy but merely an additional provocative factor among photosensitive persons. On the basis of these studies, it is believed that the attributes of patients with pattern-induced seizures—such as occurrence in adolescence and early adult life, female preponderance, and association with generalized seizure types—do not differ from those of patients with photosensitive epilepsy (4,17). To our knowledge, no study has addressed the natural history of pattern-sensitive epilepsy by long-term follow-up of a large number of well-characterized patients with pattern-induced seizures.

With these considerations, we undertook this study of 73 patients with pattern-sensitive epilepsy examined over the past half century at Mayo Clinic (Rochester, Minnesota, U.S.A.). Since 1950, when the first patient with pattern-sensitive epilepsy was evaluated (10), nearly all patients undergoing EEG examination at Mayo Clinic

who are capable of cooperating are asked to scan at least one standard pattern. This practice has enabled us to recruit the patients on the basis of their principal attribute: visual sensitivity to patterns. Furthermore, long-term follow-up data ranging from 1 to 5 decades are available for more than half of the patients.

Our objectives were threefold: (a) to define the electroclinical spectrum of pattern sensitivity, (b) to evaluate the long-term seizure and quality-of-life outcomes of patients with pattern-sensitive epilepsy, and (c) to examine whether sufficient grounds exist to consider persons with pattern-sensitive epilepsy as a distinct subgroup of those with visually induced seizures.

PATIENTS AND METHODS

Patient selection

This study was approved by the Mayo Foundation Institutional Review Board, and prior written consent to review the medical records and to gather follow-up information was obtained from each of the participants. During the 50-year period from 1950 through 1999, 79 patients had the diagnosis of pattern-sensitive epilepsy made by a Mayo Clinic neurologist. Six patients were excluded from this study: three because of failure to elicit pattern sensitivity in the EEG laboratory, and three because of failure to give consent for participation. The remaining 73 patients included in the study exhibited epileptiform discharges in the scalp EEG during pattern viewing. Patients evaluated during the 1950s and 1960s have been mentioned briefly in earlier reports (2,3,10).

Patient evaluation

All patients had scalp EEG evaluation with eight-channel recordings in the 1950s and 16 or more channels since that time. The earliest recordings were made with 14 electrodes, and all others were with 21 electrodes placed according to the 10-20 system of electrode placement. Since 1951, the standard EEG recording at Mayo Clinic has included photic stimulation and pattern testing.

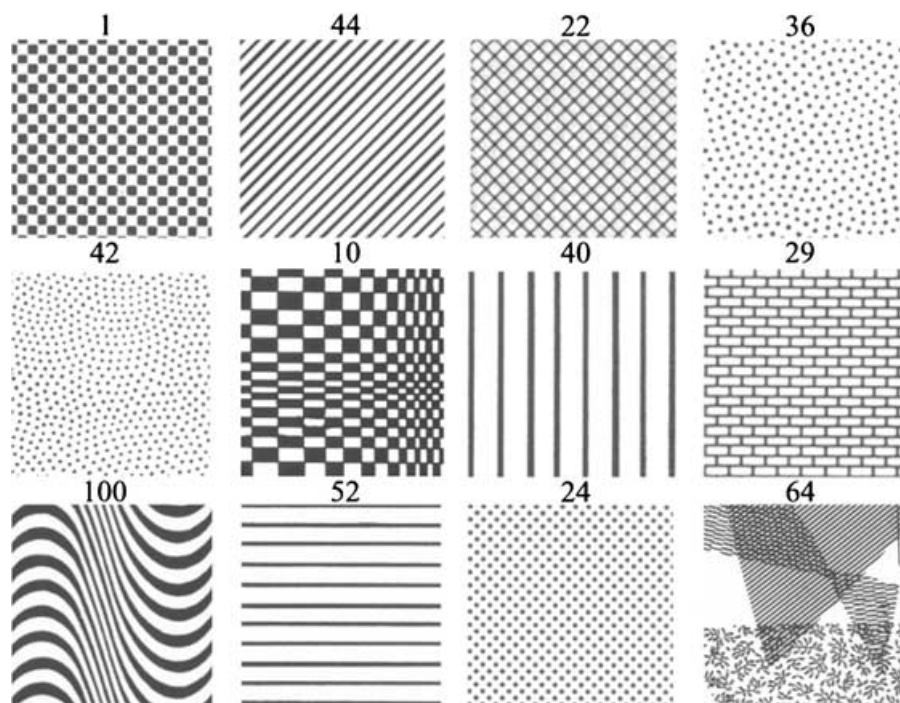
Intermittent photic stimulation

Routine photic stimulation was done first with the patient's eyes open at the programmed frequencies of 1, 3, 8, 10, 12, 15, 20, and 30 Hz. If an abnormal discharge was observed, the entire frequency spectrum from 1 through 30 was tested. After completing all programmed frequencies with the eyes open, the patient was instructed to keep the eyes closed, and the frequencies were repeated.

Pattern-sensitivity testing

Routine pattern testing was done on every patient who was capable of seeing the pattern and cooperating during the procedure. Patients were asked to scan a pattern of parallel black lines (pattern 44, Fig. 1) on an 8.5 × 11.5 inch (22 × 29 cm) laminated card for 10 s at a distance in clear focus for reading. This pattern was selected

FIG. 1. Schematic diagrams of 12 patterns used for routine testing. Each pattern is identified by the designated number.



because of its effectiveness in eliciting paroxysmal EEG discharges from a series of 100 patterns that had been tested in the first patient (2,3). Testing was conducted with the patient seated in a chair and with full-room illumination. If a paroxysmal response occurred in the EEG, the same pattern was presented repeatedly to confirm the reproducibility of EEG activation. Selected patients who exhibited a paroxysmal EEG response to the screening pattern were asked to scan 12 black-and-white photographs [8 × 8 inches (20 × 20 cm)] of other geometric patterns (Fig. 1) for 10 s each, alternating with a blank white card. Patterns that evoked an abnormality were presented again to confirm the reproducibility of EEG activation. After all the patterns were reviewed in a stationary position, they were presented again but with each pattern shaken horizontally and vertically for ~10 s.

Response testing

To assess responsiveness and reaction time during EEG paroxysms elicited during intermittent photic stimulation and pattern testing, patients were instructed to press a switch as quickly as possible after hearing an auditory click signal generated by the technician. The patients were observed closely during these procedures for any clinical manifestation of seizures and were asked to report any symptom they experienced. If a paroxysmal discharge was observed, the stimulus was interrupted to avoid any chance of precipitating a seizure.

Study definitions

We categorized seizures as proposed by the Commission on Classification and Terminology of the Interna-

tional League Against Epilepsy (ILAE) (19) and epileptic syndromes according to the revised ILAE classification (20). We used the definitions of the Committee on Terminology of the International Federation of Societies for EEG and Clinical Neurophysiology (21) to delineate epileptiform abnormalities. The interictal spontaneous epileptiform abnormalities were defined as “generalized” or “focal” according to their distribution. We defined a “definite photoparoxysmal response” as a generalized spike-wave or polyspike-wave paroxysm occurring at least twice during the same frequency of intermittent photic stimulation and a “paroxysmal pattern response” as occurring at least twice when viewing the same pattern (2,3) (Fig. 2). In addition, definitely epileptiform abnormalities confined to the posterior head region during pattern stimulation were considered significant.

Data collection

Demographic, clinical, and EEG data were ascertained through a review of medical records. Movies (16 mm) or video-EEG recordings made during the provocative procedures and original EEG records were reviewed for verification of seizures and EEG findings. We contacted the patients or their spouses, siblings, or children by letter or telephone to obtain the latest outcome results and to secure family, educational, and occupational histories.

Statistical methods

We used the mean ± standard deviation to define the dispersion. Associations of categorical variables with seizure remission rate were assessed by using χ^2 tests or Fisher’s exact test as appropriate. For continuous variables,

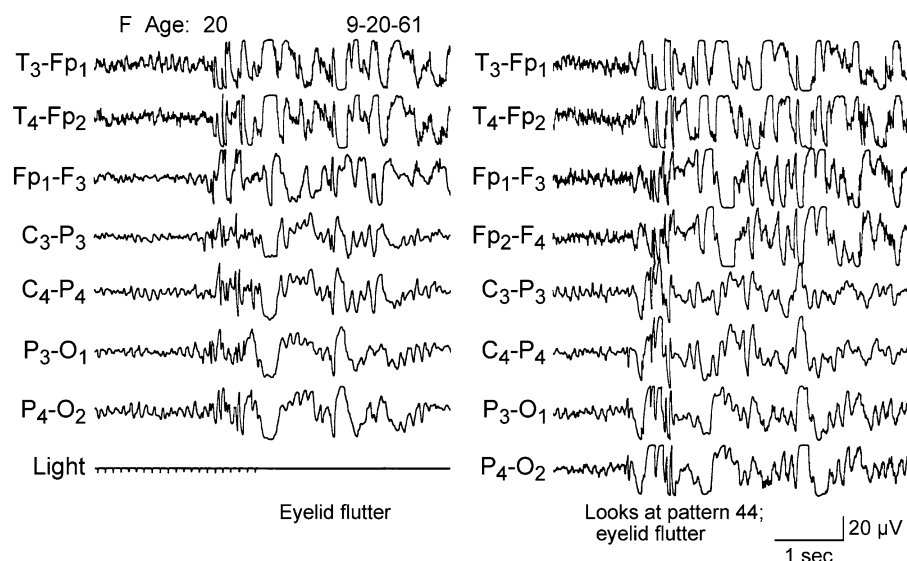


FIG. 2. Generalized paroxysmal spike-wave activity induced by photic stimulation and by viewing pattern in the same patient.

associations were evaluated using two-sample *t* tests. In the case of skewed distributions, comparisons were made by using Wilcoxon rank sum tests. To evaluate the longitudinal trend of remission, we constructed a Kaplan–Meier plot of time (22) from the date of the first seizure to the date of the last seizure. Subjects were censored at their date of last follow-up by any means. We defined “remission” as seizure free for ≥ 5 years, regardless of anticonvulsant drug (AED) treatment status (23).

RESULTS

Demographic data

The 73 patients with pattern-sensitive epilepsy included 43 female and 30 male patients (female/male ratio = 1.4:1). Thirty-one of these patients were evaluated from 1950 through 1974 and 42 from 1975 through 1999. The mean age at onset of seizures was 12.6 ± 7.1 years (median, 12.8 years; range, 0.6–32.9 years). The distribution of patients according to age at onset of seizures is shown in Fig. 3. The mean age at diagnosis was 15.1 ± 7.8 years (median, 15.9 years; range, 0.6–37.6 years). When last contacted (death or June 30, 2000), they had a mean age of 37.2 ± 12.8 years (median, 36.3 years; range, 12.0–72.4 years). The median age at diagnosis of the eight patients (six males, two females) without photosensitivity was 11.3 years (range, 4.9–16.5 years).

Clinical seizure characteristics

Types of seizures

At presentation, 60 (82.2%) patients had exhibited one or more generalized tonic–clonic seizures. The types and combinations of the seizures that occurred are listed in Table 1. The majority of patients exhibited absence, myoclonic, or generalized tonic–clonic seizures in various combinations.

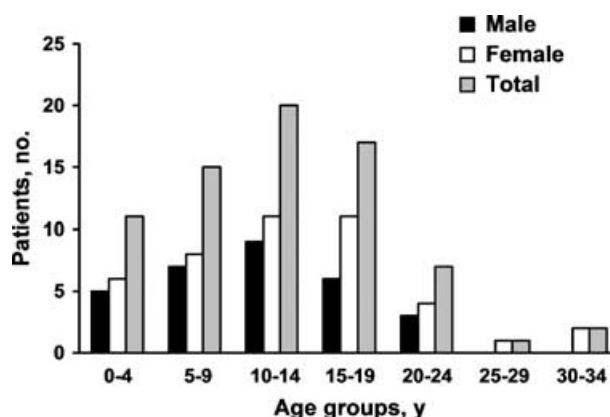


FIG. 3. Distribution of patients according to age at onset of seizures.

TABLE 1. Types of seizures in 73 patients with pattern-sensitive epilepsy

	Patients	
	No.	%
Type of seizure ever had		
Generalized tonic–clonic seizure (GTCS)	60	82.2
Absence seizure (AS)	36	49.3
Myoclonic seizure (MS)	28	38.4
Partial seizure (PS)	8	11.0
Combination of seizure types		
AS + GTCS	20	27.4
AS + MS + GTCS	19	26.0
MS + GTCS	14	19.2
PS + secondary GTCS	4	5.5
Predominant seizure type at presentation		
AS	48	65.8
MS	14	19.2
GTCS	9	12.3
PS	2	2.7

Precipitating factors

Television was the most common precipitant, being ascertained in 30 (41.1%) patients. Twenty-nine (39.7%) patients gave a clear history of one or more seizures precipitated while viewing environmental patterns such as window screens, garments, tablecloths, or ceiling tiles. Some patients initially admitted on inquiry that they felt "uncomfortable" while looking at striped patterns and, hence, preferred to avoid them, but they could not clearly state whether their seizures were related to viewing these patterns. Video game-induced seizures occurred in five (6.8%) patients. A history of seizures related to flashing lights was elicited from 28 (38.4%) patients, including three patients with seizures precipitated at discos. Five patients self-induced their seizures by compulsive hand waving against bright sunlight or by intentionally looking at patterns. When one of these five patients, the first case seen at Mayo Clinic, was examined in 1959 (9 years after his initial evaluation), he would still seek out patterns such as window screens to induce his seizures, and when placed in an environment without window screens, he would gaze at the pores of his skin. This behavior gradually abated and disappeared by age 20 years.

EEG data

Twenty-nine patients had a single EEG examination, 18 had two, and the rest had three or more EEG recordings on different occasions. The EEG findings are summarized in Table 2. Spontaneous epileptiform discharges during wakefulness or sleep were detected in 61 (83.6%) patients and were generalized in 54 (74%). Of the eight patients (11%) without photoparoxysmal responses, six had two or more EEG recordings on different occasions.

All patients exhibited epileptiform discharges to pattern stimulation, and this discharge was generalized in more than two-thirds of them. A majority of the patients were responsive to more than one pattern. The epilep-

tiform discharges during pattern viewing were similar to those during intermittent photic stimulation (Fig. 2); however, the discharges during pattern stimulation, when restricted to the posterior head region, were projected more to the parietal than to the occipital electrodes. Spontaneous fluctuations in the threshold of pattern sensitivity were occasionally noticed (Fig. 4).

In 21 (28.8%) patients, ictal clinical phenomena occurred during pattern stimulation. The predominant seizure type related to pattern viewing was brief absences, often associated with eye blinking, sursumversion, facial contractions, and myoclonic jerks of the upper extremities. Twenty-five (34.2%) patients reported subjective symptoms such as ocular discomfort, dizziness, and headache during pattern stimulation. In this retrospective study, patients were not regularly tested for television and video-game sensitivity.

Family history of seizures

Of 71 patients for whom the family history regarding seizures (exclusive of febrile seizures) was known, 15 (21%) had a first-degree relative (siblings of 10 and parents of five) with epilepsy. The mother and sister of a male patient had a history suggestive of photosensitive seizures, and one female patient had a sister with photosensitive and pattern-sensitive seizures. Other seizure types encountered were absences and generalized tonic-clonic seizures. We could not undertake a clinical and EEG evaluation of the affected family members.

Associated epileptic syndromes

All except seven patients had generalized epilepsies and epileptic syndromes (Table 3). No patient had the diagnosis of idiopathic localization-related childhood epilepsy syndrome. Among the 61 patients with idiopathic generalized epilepsies, 14 fulfilled the diagnostic criteria for juvenile myoclonic epilepsy and three had childhood

TABLE 2. EEG findings in 73 patients with pattern-sensitive epilepsy

EEG finding	Patients	
	No.	%
Interictal spontaneous epileptiform discharges		
None	12	16.4
Generalized	54	74.0
Bioccipital	2	2.7
Frontal	1	1.4
Frontocentral	1	1.4
Central	3	4.1
Photoparoxysmal response		
Absent	8	11.0
Generalized	56	76.7
Occipital	9	12.3
Pattern sensitivity		
Generalized	47	64.4
Occipitoparietal	26	35.6

TABLE 3. Distribution of 73 patients with pattern-sensitive seizures, according to epileptic syndromic diagnosis

Epilepsies and syndromes	Patients		
	No.	% of total	% of subgroup
Localization-related	7	9.6	
Cryptogenic/symptomatic	7	9.6	100
Generalized	66	90.4	
Idiopathic	59	80.8	89.4
Childhood absence epilepsy	3	4.1	5.1
Juvenile myoclonic epilepsy	14	19.2	23.7
Symptomatic	7	9.6	
Progressive myoclonus epilepsies	3	4.1	42.9
Progressive familial cerebellar ataxia with myoclonus	2	2.7	28.6
Severe myoclonic epilepsy of infancy	1	1.4	14.3
Parkinsonism-plus syndrome	1	1.4	14.3

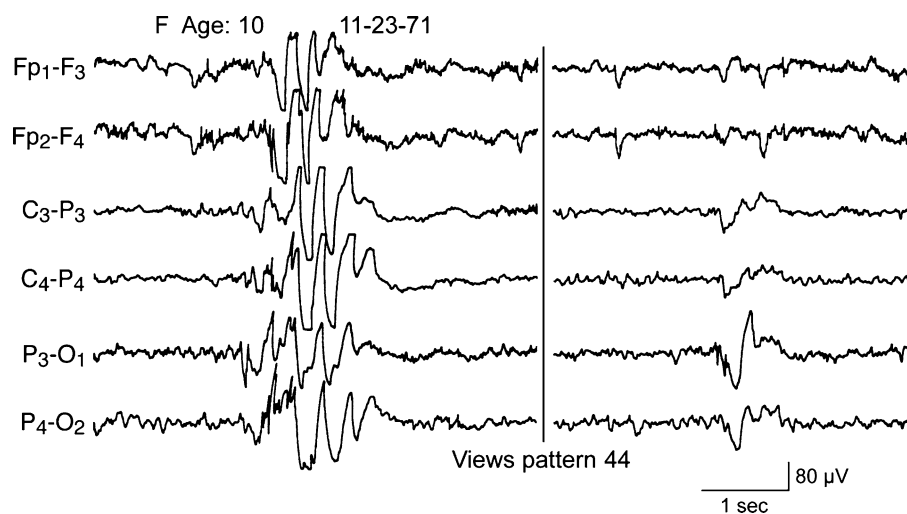


FIG. 4. Varied responses in a 10-year-old girl viewing the same pattern. Note generalized spike-wave discharge evoked in the segment on the left and discharge restricted to the posterior head regions evoked shortly afterward in segment on the right.

absence epilepsy. One male patient volunteered the symptom of reading-evoked jaw myoclonus; however, in the EEG laboratory, no epileptiform abnormalities could be elicited while the patient read. All eight patients without photosensitivity had idiopathic generalized epilepsies. In the symptomatic generalized epileptic syndromes, the clinical and EEG features suggested progressive myoclonus epilepsy in three patients, progressive familial cerebellar ataxia with myoclonus in two, and severe myoclonus epilepsy of infancy (24) in one. One male patient had a slowly progressive neurologic disease with rigidity, dystonia, bradykinesia, dementia, and supranuclear eye movement disturbance (parkinsonism-plus syndrome) that started at age 22 years. He was moderately disabled when last contacted at age 50 years. A diagnosis of mitochondrial encephalopathy was strongly suspected in another male patient who had repeated strokelike episodes. Other interesting associations of uncertain significance noted in one patient each were right temporal lobe astrocytoma, primary hyperoxaluria with renal failure, congenital biliary atresia with chronic liver disease, and pseudohypoparathyroidism.

Anticonvulsant medication

Three patients had never taken any AEDs, and five discontinued taking drugs after a short period because they did not consider them necessary. Those who were aware of the special visual stimuli that provoked their seizures were encouraged to avoid them. Phenobarbital (PB; 37 patients), sodium valproate (VPA; 34 patients), phenytoin (PHT; 31 patients), carbamazepine (CBZ; 14 patients), and primidone (PRM; 12 patients), either alone or more often in combination, constituted the AED therapy. A majority of the patients were taking AEDs prescribed by their local physicians, and the AED use profile reflected the practice that existed at different times during the study period spanning 50 years. None of the patients with a

diagnosis of progressive myoclonus epilepsy was taking PHT or CBZ when last contacted.

Follow-up data

Four patients were seen only once at Mayo Clinic, and neither they nor their relatives could be traced. The mean follow-up for the other 69 patients was 16.6 ± 12.5 years (median, 15.7 years; range, 3 months–49.6 years). Fifty-five patients were followed up for ≥ 5 years, 47 for ≥ 10 years, and 17 for ≥ 25 years. Eleven patients died. Two deaths were seizure related, and the others were due to a progressive neurologic or systemic disease or an uncertain cause.

Seizure outcome

Of the 55 patients who were followed up for ≥ 5 years, 25 (45.5%) had achieved complete remission of seizures for ≥ 5 years, and nine of them were no longer taking AEDs. After having had a few initial seizures, 15% of the patients became seizure free within 1 year after seizure onset, but the rest did not have remission for several years (Fig. 5). The proportion of patients in remission at 5 and

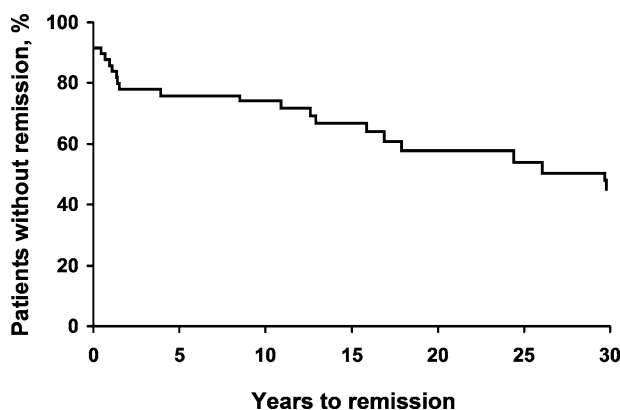


FIG. 5. Kaplan-Meier plot of time for 53 patients from first seizure to remission or latest follow-up. Two patients were excluded because of incomplete information.

TABLE 4. Responses of 58 patients with pattern-sensitive epilepsy to quality-of-life-related questions

	Patients who answered "No"	
	No.	%
Were you disabled by seizures?	41	70.7
Did epilepsy affect your education?	42	72.4
Did epilepsy affect your occupation?	40	69.0
Did epilepsy affect your family life?	46	79.3

TABLE 5. Patient attributes significantly associated with seizure remission

Attribute	Remission rate, no. of patients (%)	p Value ^a
Family history of seizures	Present 9/13 (69.2) Absent 15/41 (36.6)	0.06
Progressive neurologic disease	Present 0/7 (0.0) Absent 25/47 (53.2)	0.01
Seizure-related disability	Present 1/16 (6.3) Absent 23/37 (62.2)	0.001
Educational underachievement	Present 0/16 (0.0) Absent 24/38 (63.2)	0.001
Occupational underachievement	Present 1/20 (5.0) Absent 23/34 (67.6)	0.001
Seizure-related disturbed family life	Present 0/12 (0.0) Absent 24/42 (57.1)	0.001

^aFisher's exact test.

10 years after seizure onset was 25%, and at 15 and 20 years, 34% and 42%, respectively. The mean age at remission was 26.4 ± 12.9 years (median, 24.4 years).

Quality-of-life outcome

The distribution of the responses for 58 patients who were available to answer the quality-of-life-related questions is provided in Table 4. More than two-thirds of the patients admitted they were not disabled by seizures and did not think that epilepsy affected their educational achievement, occupational status, and family life.

Factors that influenced outcome

Among the factors examined in relation to remission, absence of progressive neurologic disease was significantly associated with remission (Table 5). A family history of seizures showed a trend in favor of remission. No relation was noted between seizure remission and the following variables: age at seizure onset, sex, seizure type or types, or precipitating factor or factors. As expected, patients with remission of seizures achieved better educational and occupational outcomes and were less likely to report feeling disabled by their seizures.

DISCUSSION

To our knowledge, this is the first study that has assembled a sizeable number of patients with pattern-

sensitive epilepsy on the basis of their principal attributes: pattern-induced epileptiform abnormalities in the EEG and seizures. We think that our study overcame the ascertainment bias that has complicated previous studies (4,18,25), which selected patients with pattern sensitivity from among those with photosensitivity. Additional strengths of our study include a standardized clinical and EEG evaluation and long follow-up, which has helped us to define for the first time the natural history of pattern-sensitive epilepsy. We acknowledge that although we were able to gather the latest information about the seizure and quality-of-life outcomes of the majority of patients through postal or telephone interviews, we were unable to reexamine them electroencephalographically.

Demographic features

The median age (13 years) at seizure onset of our cohort of patients with pattern-sensitive epilepsy was similar to that of those with photosensitive epilepsies in general (17,18,26), and with television- (6,27) and video game-induced seizures (28–30) in particular. Although nearly two thirds of patients with photosensitive seizures are females (17,18), the female preponderance appears to be less pronounced in the subgroup with pattern sensitivity. The female-to-male ratio in our cohort was 1.4:1. It is interesting to note that among our eight patients without photosensitivity, six were males. In a comparison of the sex distribution of a group of 11 patients (seven males and four females) with pattern sensitivity to 34 patients (13 males and 21 females) with pattern sensitivity and photosensitivity, Brinciotti et al. (16) noted a male preponderance for those with pattern sensitivity. A male preponderance has been documented consistently in video-game epilepsy; nearly three-fourths of patients in this subgroup of visually induced seizures are males (28,29,31). Although this has been attributed to a greater number of boys than girls playing video games (28,32), a survey of American school children by Funk (33) found that although girls were less involved in arcade play, a majority of them regularly played home video games. The difference in the sex distribution in the subgroups of patients with visually induced seizures may be an expression of the genotypic and phenotypic variability among these disorders.

Environmental triggering factors

At least 80% of patients with photosensitive epilepsy volunteer a history of seizures precipitated by environmental visual stimuli (17). By contrast, in different studies, direct questioning implicated pattern as a seizure trigger in only 6% (18) to 31% (14) of pattern-sensitive patients. At presentation, a majority of our patients were unaware of their seizure trigger, but on specific inquiry, they all admitted that they preferred to avoid looking at patterned objects because these objects made them uncomfortable. Furthermore, environmental stimuli may go unnoticed by the patient, either because the stimulus is too mild to

produce a clinical seizure or the relation between the stimulus and the response is too variable. For example, one of our female patients with no clinical history of pattern sensitivity was discovered incidentally to have epileptiform EEG discharges when she gazed at a row of jacketed EEG recordings that had been filed upright on a shelf in the EEG recording room.

A wide variety of environmental visual patterns such as window screens, patterned clothing, ceiling tiles, folding doors, shutters, telephone lines against a bright sky, and the grating of escalator steps have been implicated in triggering seizures in pattern-sensitive patients (4,5). The first patient with this seen at Mayo Clinic could even self-induce seizures by looking at the pores of his skin. However, the most frequent environmental stimulus found to induce seizures in pattern-sensitive patients, as observed in our study, is television (4–6,27).

Some patients, including five of ours, who have pattern-sensitive epilepsy actively self-induce their seizures, as do those with other forms of reflex epilepsies. This behavior is often exceedingly difficult to control. All five children with self-induced pattern-sensitive seizures reported by Matricardi et al (15) had multiple seizure types, medical refractoriness, and mental retardation. However, of our five patients who regularly self-induced their seizures, four were neurologically and mentally normal.

Seizure semiology and EEG features

The high variability observed in the distribution of seizure types and EEG findings across studies on visually induced seizures may be due to inconsistent terminology used in describing them (4,5,17). A recent proposal for terminology and classification of clinical and EEG phenomenology (34) may help in standardized ascertainment of these data and their comparison between different studies and between different visually induced seizures.

Absences, often associated with eye blinking, sursumversion, facial contractions, and myoclonic jerks of the upper extremities, are the seizures typically evoked by pattern stimulation (4,5). More than 80% of our patients had experienced one or more generalized tonic-clonic seizures by the time they were first evaluated. Partial seizures with complex visual hallucinations followed by unresponsiveness and automatisms occurred in only 11% of our patients. The seizure manifestations of pattern-sensitive epilepsy are remarkably similar to those of other visually induced reflex epilepsies (17,18).

In patients with pattern sensitivity, the most effective stimulus for triggering epileptiform abnormalities in the EEG laboratory consists of simple black-and-white geometric patterns with sharply contrasted interfaces, especially parallel lines and closely spaced dots; complex pictures with indistinct borders are generally ineffective (2–5). Change of fixation of gaze while viewing a pattern is essential for eliciting epileptiform abnormalities

(2–5). Rhythmic movement of the pattern often increases the discharges (2–5). Differences in activation based on the orientation of the pattern have been found to be important for some patients but not others (3,5,10). Pattern vision can rarely inhibit the paroxysmal discharges (35), and pattern withdrawal may rarely activate the discharges (5).

The typical EEG response to full-field pattern stimulation in pattern-sensitive patients is a generalized spike-wave discharge that usually is bisynchronous and bilaterally symmetrical (4,5) (Fig. 2). In contrast to intermittent photic stimulation, which elicits discharges that are generalized from the onset or rapidly become generalized, pattern stimulation more often results in subtly graded EEG responses confined to the posterior head regions (4,5). With hemifield stimulation, Wilkins et al. (36) demonstrated in 15 patients with pattern sensitivity that the responses were maximal contralateral to the field of stimulation.

Although most patients with pattern sensitivity also exhibit photosensitivity, pattern sensitivity does occur in isolation. Spontaneous fluctuations in the threshold for photosensitivity and pattern sensitivity are well known (Fig. 4) (3,37,38). For example, a patient may be extremely sensitive on the first day of testing but may not be sensitive to intermittent light or patterns on the following day, which makes the interpretation of EEG activation results difficult unless confirmed by repeated testing. Of our patients, eight (11%) were not photosensitive with conventional frequencies of intermittent photic stimulation, and five of them were tested two or more times on different occasions. Wilkins et al. (39) found a significant association between pattern sensitivity and sensitivity to intermittent photic stimulation frequencies >30 Hz. Because our photic stimulation protocol does not include testing at >30 Hz, we could not verify this through our retrospective study. However, other authors have found a small proportion of pattern-sensitive patients who were insensitive to intermittent photic stimulation. In a group of 67 children with visually induced seizures studied by Brincioti et al. (16), 51% exhibited sensitivity to both light and pattern, but 33% had isolated sensitivity to light and 16% to pattern. Among patients with television epilepsy and video-game epilepsy, ~20% are not responsive to intermittent light stimulation but are sensitive to patterns (6,32).

Associated epileptic syndromes

The strong association of photosensitivity with idiopathic generalized epilepsies, especially with juvenile myoclonic epilepsy, and symptomatic epilepsies such as severe myoclonic epilepsy of infancy and other progressive myoclonus epilepsies is well recognized (26,40). We observed a similar trend for pattern sensitivity. Fourteen of our patients (19.2% of the total and 23.7% of

those with idiopathic generalized epilepsies) had clinical and EEG findings consistent with juvenile myoclonic epilepsy. Of 103 patients with photosensitivity studied by Wolf and Goosses (26), one-third had juvenile myoclonic epilepsy. In a series of 35 patients with video game-induced seizures, one-third of the patients with idiopathic generalized epilepsy had juvenile myoclonic epilepsy (41).

Clinical genetics

As in photosensitivity, genetic factors have long been suspected to be important in pattern sensitivity. Two of the four patients with pattern sensitivity studied by Chatrian et al. (13) were brothers. Brinciotti et al. (42) reported two families in which five members (three in one and two in the other) had pattern-sensitive epilepsy. In a group of children with visually induced seizures, two girls who had both photosensitivity and pattern sensitivity were monozygotic twins (16). Fifteen (21%) of our patients had one or more first-degree relatives with idiopathic generalized epilepsies. Photosensitive or pattern-sensitive epilepsy (or both) occurred in two families. Many of the epileptic syndromes with which pattern sensitivity is associated, such as juvenile myoclonic epilepsy, childhood absence epilepsy, and progressive myoclonus epilepsies, have themselves a strong genetic basis (43). More than one-third of patients with severe myoclonic epilepsy of infancy may exhibit photosensitivity and pattern sensitivity at an age as young as 3 months (24), as happened in one of our patients.

Natural history

The natural history of photosensitivity has been well studied. Harding et al. (44) followed up 100 patients with photosensitivity for an average of 14 years and a mean age at last follow-up of 27 years. Seventy-seven patients became seizure free. Photosensitivity as tested in the EEG laboratory persisted in nearly two-thirds of patients. In the other one-third, it disappeared at about age 24 years.

Little is known about the natural course of pattern-sensitive epilepsy and how it affects a patient's educational and occupational performance. The present study appears to be the first to inquire into the long-term seizure and quality-of-life outcomes of a sizeable number of patients with pattern-sensitive epilepsy followed up for several decades. Of our 55 patients who had ≥ 5 years of follow-up, 25 (45.5%) achieved remission, defined as seizure free for ≥ 5 years, irrespective of AED treatment status. Remission occurred at a median age of 24 years, after patients had had seizures for several years. Absence of progressive neurologic disease was significantly correlated with remission. A family history of seizures (denoting idiopathic generalized epilepsy syndromes) showed a trend in favor of remission.

Pathophysiology of visually induced seizures

Discoveries in the past 3 decades have changed our conception of how the visual brain functions. Anatomic and physiologic observations in monkeys indicate that in addition to the primary visual area, the primate visual cortex consists of several separate and functionally independent specialized areas that analyze different aspects of the retinal image such as color, form, orientation, and motion and possibly other attributes of the visible world (45,46). The primary visual area receives input signaling different aspects of visual stimuli and assembles and initially processes the information before relaying it to specialized visual areas (46). The relative contribution of the parvocellular and magnocellular visual systems in the transmission of different aspects of visual stimuli to visual cortex is still debated (25,46).

The role played by factors involved in the genesis of epileptogenic activity in the EEG of patients with pattern sensitivity, such as the type of pattern, size, sharpness of boundaries, luminance, degree of contrast, orientation and movement, the effects of monocular and binocular stimulation, selective stimulation of the visual fields, and influence of AEDs, has been investigated extensively (2,3,13,14,47,48). These studies have demonstrated that the epileptiform activity evoked by patterns arises within the visual cortex. However, because of poor spatial resolution of EEG, reliable discrete localizations for different visual functions will be difficult to obtain though these lines of investigation. Moreover, a critical level of excitation anywhere within a hyperexcitable cortical region may result in generalized paroxysmal epileptiform activity, depending on the magnitude of excitation and induction of cortical-subcortical circuits (49,50).

Positron emission tomographic studies on humans showed an increase in regional cerebral blood flow in the prestriate mesial occipital region (fusiform and lingual gyri) (51) when subjects viewed a multicolored abstract painting containing no recognizable objects. However, when they viewed a pattern of moving black and white squares, the highest rate of cerebral blood flow was more lateral, in the temporoparietooccipital junction (52). Darby et al. (47) noted that the topography of the EEG activity in response to patterns, when focal, is usually maximal over the posterior temporal and parietal scalp electrodes instead of the occipital electrodes; this is consistent with the positron emission tomographic results and our observations.

Discrete excitable loci within a region such as visual cortex that are concerned with specific attributes of visual function such as color perception or pattern identification may dictate the susceptibility to a specific stimulus and the resultant clinical substrate of its excitation. Accordingly, either a single specific trigger (intermittent light, pattern, or color) or, more frequently, closely related multiple triggers (intermittent light, pattern, and color) might provoke

the seizure phenomenon. The authors encourage further research in patients who have pattern-sensitive epilepsy with positron emission tomography and functional magnetic resonance imaging to delineate the precise anatomic substrate of ictogenesis.

Delineation of pattern-sensitive epilepsy as an epileptic syndrome

A majority of EEG laboratories do not routinely perform pattern-sensitivity testing and conduct it only in persons who exhibit an abnormal response to intermittent photic stimulation. The existing notion about the almost invariable association between photosensitivity and pattern sensitivity possibly resulted from this practice. However, as shown by us and others (6,16,35), 10% to 20% of persons who are sensitive to pattern stimuli are consistently unresponsive to the conventional frequency range of intermittent photic stimulation. Among our eight patients without photosensitivity, six were males, and all had idiopathic generalized epilepsies. As described earlier, it is plausible pathophysiologically to have selective vulnerability to a specific visual stimulus such as light, pattern, or color.

Although patients with pattern-sensitive epilepsy share most of the characteristics of those with photosensitive epilepsy, subtle differences exist between these two epilepsy syndromes. In a comparative study of the clinical and EEG findings in children categorized according to their sensitivity to light or pattern (or both), Brinciotti et al. (16) noted a higher occurrence of localization-related symptomatic epilepsies, neurologic abnormalities, and focal EEG abnormalities in those with pattern sensitivity without photosensitivity. In our group of patients with pattern-sensitive epilepsy, we observed a less convincing female preponderance and more graded EEG responses with a different topography compared with those with photosensitive epilepsy. For these reasons, it seems reasonable to regard pattern-sensitive epilepsy as a distinct subtype of visually induced reflex epilepsy.

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